

Hospital Management System (HMS)

Project Title



<Student Name (Roll No)>: **Muhammad Hassan (F22BINFT1E02090)**

Submitted to: **Dr. Rana Nazeer sb**

Department of Information Technology

The Islamia University of Bahawalpur

Software Design Document (SDD)

Project Title

Hospital Management System (HMS)

Document Control

Revision History

Version	Date	Description	Author
1.0	06/ 08/ 2026	Initial Submission	Muhammad Hassan

Table of Contents

- 1. Introduction**
- 2. System Overview**
- 3. Design Considerations**
- 4. System Architecture**
- 5. Data Design**
- 6. Component Design**
- 7. Interface Design**
- 8. Security Design**
- 9. Performance & Reliability**
- 10. Error Handling & Logging**
- 11. Assumptions & Constraints**
- 12. Future Enhancements**

13. Conclusion

14. References

1. Introduction

1.1 Purpose

This Software Design Document (SDD) describes the complete design of the Hospital Management System (HMS). It explains the system architecture, data structures, modules, interfaces, and design decisions based on the approved Software Requirement Specification (SRS).

1.2 Scope

This document is intended for project supervisors, evaluators, and developers to understand how the system will be implemented and structured.

1.3 Definitions & Acronyms

- HMS: Hospital Management System**
 - SRS: Software Requirement Specification**
 - SDD: Software Design Document**
 - DBMS: Database Management System**
-

2. System Overview

The Hospital Management System is a web-based application designed to automate hospital operations such as patient registration, appointment scheduling, billing, pharmacy management, and laboratory reporting. The system provides role-based access to ensure secure and efficient usage.

3. Design Considerations

3.1 Design Goals

- **Modular and scalable architecture**
- **Easy-to-use interface**
- **Secure data handling**
- **Maintainability and extensibility**

3.2 Constraints

- **Limited development time**
- **Academic project scope**
- **Requires internet or local server environment**

3.3 Assumptions

- **Users have basic computer knowledge**

- Hospital staff will be trained before usage
-

4. System Architecture

4.1 Architectural Style

The system follows a three-tier architecture: 1. Presentation Layer (Frontend) 2. Application Layer (Backend) 3. Data Layer (Database)

4.2 Architecture Diagram

(To be designed using Draw.io or similar tool)

5. Data Design

5.1 Database Overview

The database is designed using a relational model with MySQL.

5.2 Major Tables

Patient

- patient_id (PK)
- name
- age
- gender
- contact
- address

Doctor

- doctor_id (PK)
- name

- specialization
- availability

Appointment

- appointment_id (PK)
- patient_id (FK)
- doctor_id (FK)
- date_time
- status

Billing

- bill_id (PK)
- patient_id (FK)
- total_amount
- payment_status

Medicine

- medicine_id (PK)
- name
- price
- quantity

LabTest

- test_id (PK)
- patient_id (FK)
- test_name
- report

Users

- user_id (PK)
- username
- password
- role

5.3 ER Diagram (Entity Relationship Diagram)

The ER Diagram represents the logical structure of the Hospital Management System database and shows the relationships among different entities.

Entities and Attributes

Patient

- patient_id (PK)
- name
- age
- gender
- contact_no
- address

Doctor

- doctor_id (PK)
- name
- specialization
- contact_no

Appointment

- appointment_id (PK)
- appointment_date
- appointment_time
- status
- patient_id (FK)
- doctor_id (FK)

Billing

- bill_id (PK)
- bill_date
- total_amount

- payment_status
- patient_id (FK)

Medicine

- medicine_id (PK)
- medicine_name
- price
- quantity

LabTest

- test_id (PK)
- test_name
- result
- patient_id (FK)

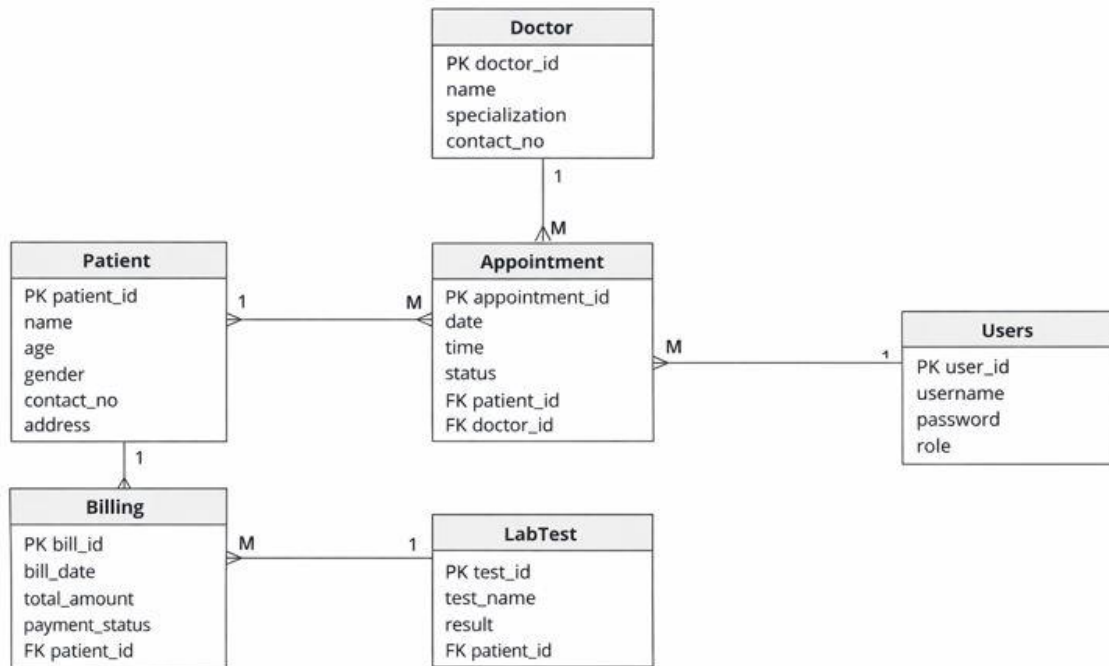
Users

- user_id (PK)
- username
- password
- role

Relationships

- One Patient can have *many* Appointments (1:M)
- One Doctor can attend *many* Appointments (1:M)
- One Patient can have *many* Bills (1:M)
- One Patient can have *many* Lab Tests (1:M)

- **Users table manages authentication for Admin, Doctor, Receptionist, Pharmacist, and Lab Technician**



6. Component Design

6.1 Patient Management Module

Handles patient registration, updates, search, and history maintenance.

6.2 Doctor Management Module

Manages doctor profiles, schedules, and availability.

6.3 Appointment Module

Responsible for booking, updating, and canceling appointments.

6.4 Billing Module

Generates bills, calculates charges, and manages payments.

6.5 Pharmacy Module

Manages medicine inventory, stock updates, and invoicing.

6.6 Laboratory Module

Handles test requests, report uploads, and patient lab history.

6.7 Admin Module

Provides full system control including user management and reports.

7. Interface Design

7.1 User Interface

- **Login Page**
- **Dashboard (Role-based)**
- **Forms for data entry**

- **Tables for record display**

7.2 Hardware Interface

- **Standard PC or Laptop**
- **Server/Localhost environment**

7.3 Software Interface

- **MySQL Database**
 - **Web Server (XAMPP/WAMP)**
-

8. Security Design

- **Authentication using username and password**
 - **Role-based authorization**
 - **Secure database access**
 - **Restricted admin privileges**
-

9. Performance & Reliability

- **Supports multiple users simultaneously**

- **Fast response time (2–3 seconds)**
 - **Reliable data storage and retrieval**
-

10. Error Handling & Logging

- **Input validation at frontend and backend**
 - **User-friendly error messages**
 - **System logs for debugging**
-

11. Assumptions & Constraints

- **System will be used in small to medium hospitals**
 - **Internet connectivity required for online deployment**
-

12. Future Enhancements

- **Mobile application support**
- **Online payment gateway**
- **SMS/Email notifications**

- OPD & IPD modules
 - Ambulance management
-

13. Conclusion

The Software Design Document provides a detailed design blueprint for the Hospital Management System. The proposed design ensures scalability, security, and efficiency while fulfilling all requirements mentioned in the SRS.

14. References

- Sommerville, I. *Software Engineering*
- Pressman, R. *Software Engineering: A Practitioner's Approach*
- MySQL Documentation
- MDN Web Docs